

# Immunogenicity and expression - Fig6

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## Warning: le package 'knitr' a été compilé avec la version R 4.4.3

## Load libraries

```
library(dplyr)
library(tidyr)
library(ggplot2)
library(ggpubr)
library(ggsignif)
```

## Load and prepare data

```
# Read immunogenicity table
df_imgg <- read.table("../data/peptide_table_imgg_pred.tsv",
  sep = "\t", header = T)
df_imgg <- df_imgg[df_imgg$allele %in% "HLA-A*02:01", ]

# Read CTA peptides
df_cta_pep <- read.table("../results/peptides_CTA_uniques_jj_ch_iedb.tsv",
  sep = "\t", header = T)
colnames(df_cta_pep) <- c("Gene", "peptide")
df_imgg_genes <- merge(df_imgg, df_cta_pep, by = "peptide", all.x = T)

# Prepare scores df
df_gene_score <- df_imgg_genes %>%
  group_by(Gene) %>%
  summarise(mean_score = mean(immunogenicity.score, na.rm = TRUE),
    sum_score = sum(immunogenicity.score, na.rm = TRUE),
    n_peptides = n()) %>%
  arrange(desc(sum_score))

# Read CTAs
l_cta <- read.table("../data/CTA_list_clean.txt", header = F)$V1

# Read immunophenotype data
df_immp <- read.table("../results/clusters_cta_pearson_63.tsv",
  sep = "\t", header = T)
colnames(df_immp) <- c("Cluster", "Gene")

# Merge immunogenicity scores with immunophenotype
df_imgg_immp <- merge(df_gene_score, df_immp, by = "Gene", all.x = T)

# Cluster data
df_clust <- read.table("../results/clusters_cta_signif_conv_indiv.tsv",
  sep = "\t", header = T)

# Mean expression data
df_mean_expr_norm <- read.table("../results/table_scatter_63_complete.tsv",
  sep = "\t", header = T)
```

## I. Boxplot immunogenicity by hot and cold clusters

```
# Select CTAs with peptides
df_imgg_imp_h_c <- df_imgg_imp[!is.na(df_imgg_imp$Cluster) &
  !is.na(df_imgg_imp$mean_score), ]

# Verify normality
shapiro.test(df_imgg_imp_h_c$mean_score)

##
## Shapiro-Wilk normality test
##
## data: df_imgg_imp_h_c$mean_score
## W = 0.96287, p-value = 1.71e-12

shapiro.test(df_imgg_imp_h_c$mean_score)

##
## Shapiro-Wilk normality test
##
## data: df_imgg_imp_h_c$mean_score
## W = 0.96287, p-value = 1.71e-12

# Wilcoxon test
wilcox.test(mean_score ~ Cluster, data = df_imgg_imp_h_c)

##
## Wilcoxon rank sum test with continuity correction
##
## data: mean_score by Cluster
## W = 62740, p-value = 0.6433
## alternative hypothesis: true location shift is not equal to 0

# Plot
df_imgg_imp_h_c$Cluster <- factor(df_imgg_imp_h_c$Cluster,
  levels = c("Hot tumor CTAs", "Cold tumor CTAs"))
p <- ggplot(df_imgg_imp_h_c, aes(x = Cluster, y = mean_score,
  fill = Cluster)) + geom_boxplot(alpha = 0.7, outlier.shape = NA,
  width = 0.2) + geom_jitter(width = 0.12, size = 0.5, alpha = 0.8) +
  scale_fill_manual(values = c(`Cold tumor CTAs` = "#625f6aff",
    `Hot tumor CTAs` = "#9e9e9eff")) + labs(x = "", y = "Immunogenicity (mean score)") +
  stat_compare_means(comparisons = list(c("Cold tumor CTAs",
    "Hot tumor CTAs")))) + theme_classic()
p

# ggsave('.../results/boxplot_cold_hot.svg', plot = p, width
# = 15, height = 10, units = 'cm')
```

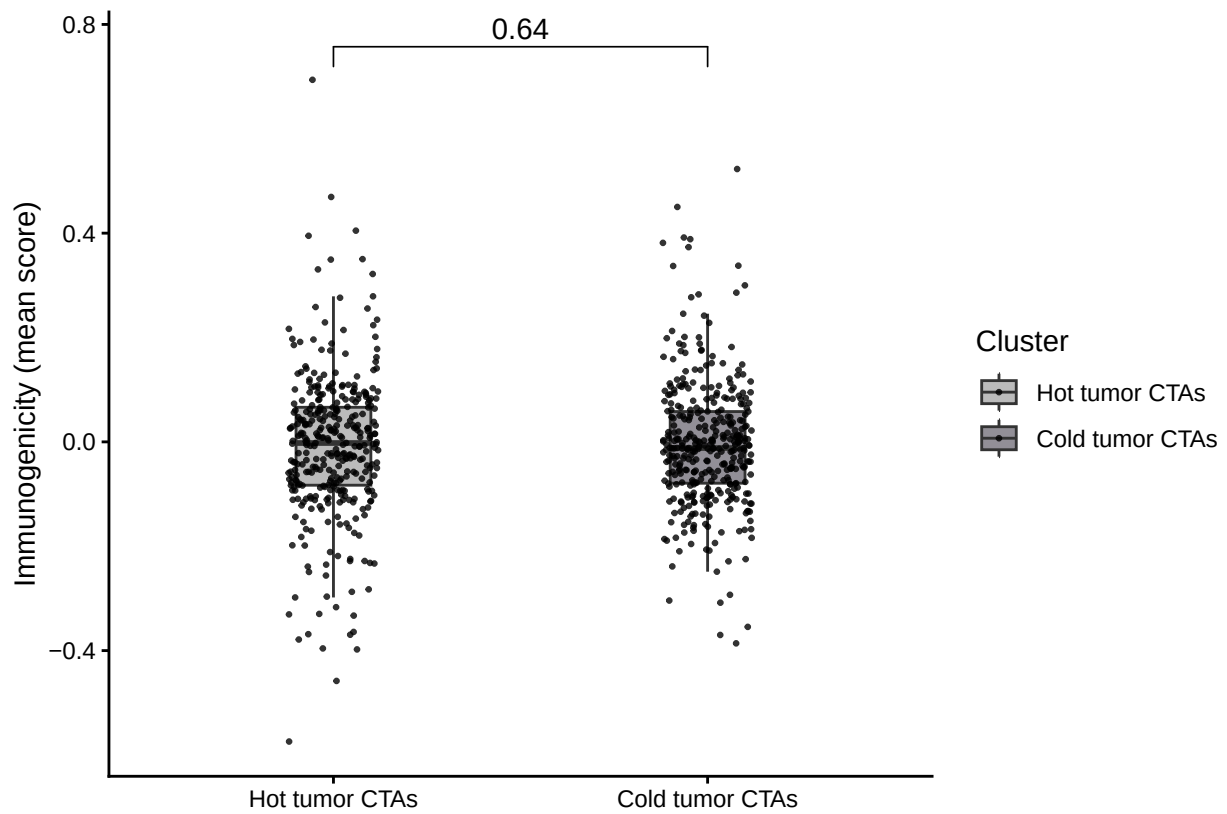


Figure 1: Boxplot immunogenicity by hot and cold clusters

## II. Boxplot immunogenicity by survival CTAs clusters

```
# Rename col and rename clusters
colnames(df_clust) <- c("Cluster_surv", "Gene")

# Merge
df_imgg_imp_clust <- merge(df_imgg_imp, df_clust, by = "Gene")
df_imgg_imp_clust$Cluster_surv <- as.character(df_imgg_imp_clust$Cluster_surv)

# Pairwise wilcoxon test with BH correction
pwt <- pairwise.wilcox.test(df_imgg_imp_clust$mean_score, df_imgg_imp_clust$Cluster_surv,
  p.adjust.method = "BH")

# Extract adjusted p-values
p_ab <- pwt$p.value["B", "A"]
p_ac <- pwt$p.value["C", "A"]
p_bc <- pwt$p.value["C", "B"]

# Convert into stars
signif_stars <- function(p) {
  if (p <= 1e-04)
    "****" else if (p <= 0.001)
    "***" else if (p <= 0.01)
    "**" else if (p <= 0.05)
    "*" else "ns"
}

# Plot
p <- ggplot(df_imgg_imp_clust, aes(x = Cluster_surv, y = mean_score,
  fill = Cluster_surv)) + geom_boxplot(alpha = 0.7, outlier.shape = NA,
  width = 0.3) + geom_jitter(width = 0.12, size = 1, alpha = 0.8) +
  labs(x = "CTAs cluster", y = "Immunogenicity (mean score)") +
  scale_fill_manual(values = c(A = "#3498DB", B = "#2ECC71",
    C = "#E74C3C")) + theme_classic() + theme(legend.position = "none") +
  geom_signif(comparisons = list(c("A", "B"), c("A", "C"),
    c("B", "C")), annotations = c(signif_stars(p_ab), signif_stars(p_ac),
    signif_stars(p_bc)), step_increase = 0.1, tip_length = 0.02)

p

# ggsave('../results/boxplot_clust_a_b_c.svg', plot = p,
# width = 15, height = 10, units = 'cm')
```

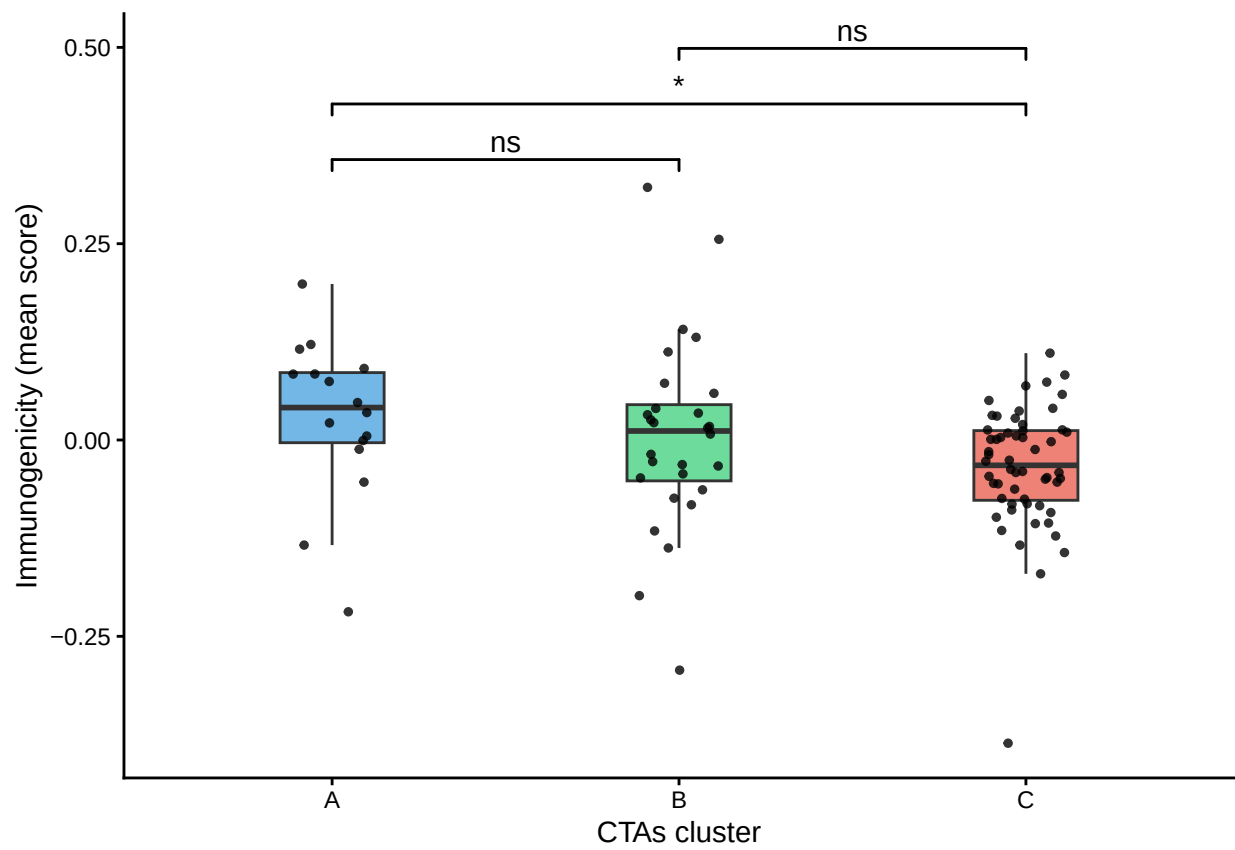


Figure 2: Boxplot immunogenicity by survival CTAs clusters



### III. Correlation normal tissues and mean score

```
# Merge
df_mean_expr_norm$Gene <- df_mean_expr_norm$SYMBOL
df_corr_means <- merge(df_imgg_imp_clust, df_mean_expr_norm,
  by = "Gene")
df_corr_means <- na.omit(df_corr_means)
rownames(df_corr_means) <- df_corr_means$Gene

# Spearman correlation
res <- cor.test(df_corr_means$mean_score, df_corr_means$Mean_expression_tissues,
  method = "spearman")

## Warning in cor.test.default(df_corr_means$mean_score,
## df_corr_means$Mean_expression_tissues, : Impossible de calculer la p-value
## exacte avec des ex-aequos

# Prepare label for the plot
r_value <- round(res$estimate, 7)
p_value <- round(res$p.value, 5)
label_text <- paste0("r = ", r_value, "\np = ", p_value)

# Plot
p <- ggplot(df_corr_means, aes(x = Mean_expression_tissues, y = mean_score,
  color = Cluster_surv)) + geom_point() + geom_smooth(method = "lm",
  se = FALSE, color = "black") + scale_color_manual(values = c(A = "#3498DB",
  B = "#2ECC71", C = "#E74C3C")) + labs(x = "CTAs expression in normal tissues (Log10 TPM mean)",
  y = "Immunogenicity (mean score)", color = "CTAs cluster") +
  annotate("text", x = max(df_corr_means$Mean_expression_tissues) *
    0.04, y = max(df_corr_means$mean_score) * 0.9, label = label_text,
    hjust = 0) + scale_x_log10() + theme_minimal()
p

## `geom_smooth()` using formula = 'y ~ x'

# ggsave('../results/plot_linear_regression_spearman_clusts.svg',
# plot = p)
```

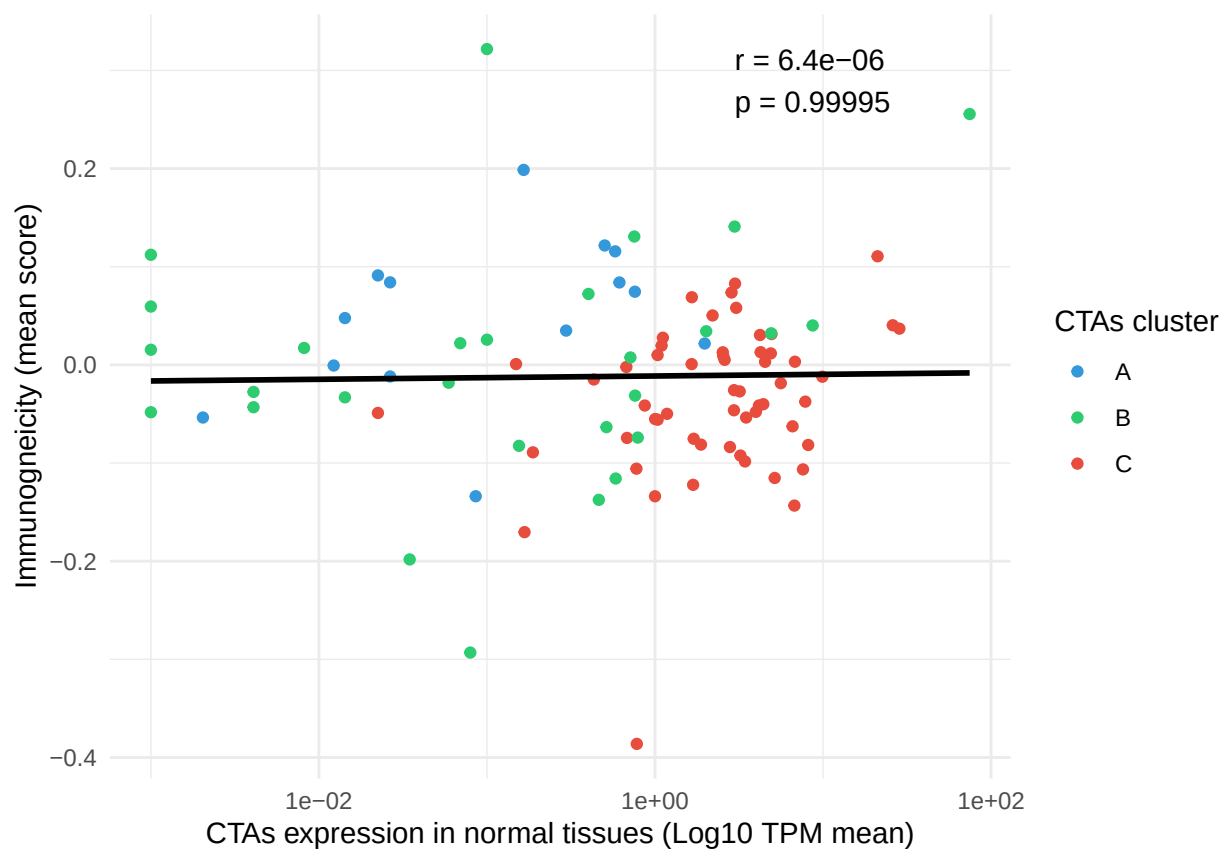


Figure 3: Correlation plot

```

# Select data
df_clean <- df_mean_expr_norm %>%
  filter(!is.na(Cluster)) %>%
  mutate(TPM_log2 = log2(Mean_expression_tissues + 1))

# Pairwise comparisons
comparisons <- combn(unique(df_clean$Cluster), 2, simplify = FALSE)

# Plot
p <- ggplot(df_clean, aes(x = Cluster, y = TPM_log2, fill = Cluster)) +
  scale_fill_manual(values = c(A = "#3498DB", B = "#2ECC71",
    C = "#E74C3C")) + stat_summary(fun = mean, geom = "bar",
  width = 0.4) + stat_summary(fun.data = mean_se, geom = "errorbar",
  width = 0.2) + theme_classic() + labs(x = "CTAs cluster",
  y = "CTAs expression in normal\ntissues (Log2(TPM + 1))")

# Add tests
plot_final <- p + stat_compare_means(comparisons = comparisons,
  method = "t.test", label = "p.signif", step.increase = 0.1)
plot_final

```

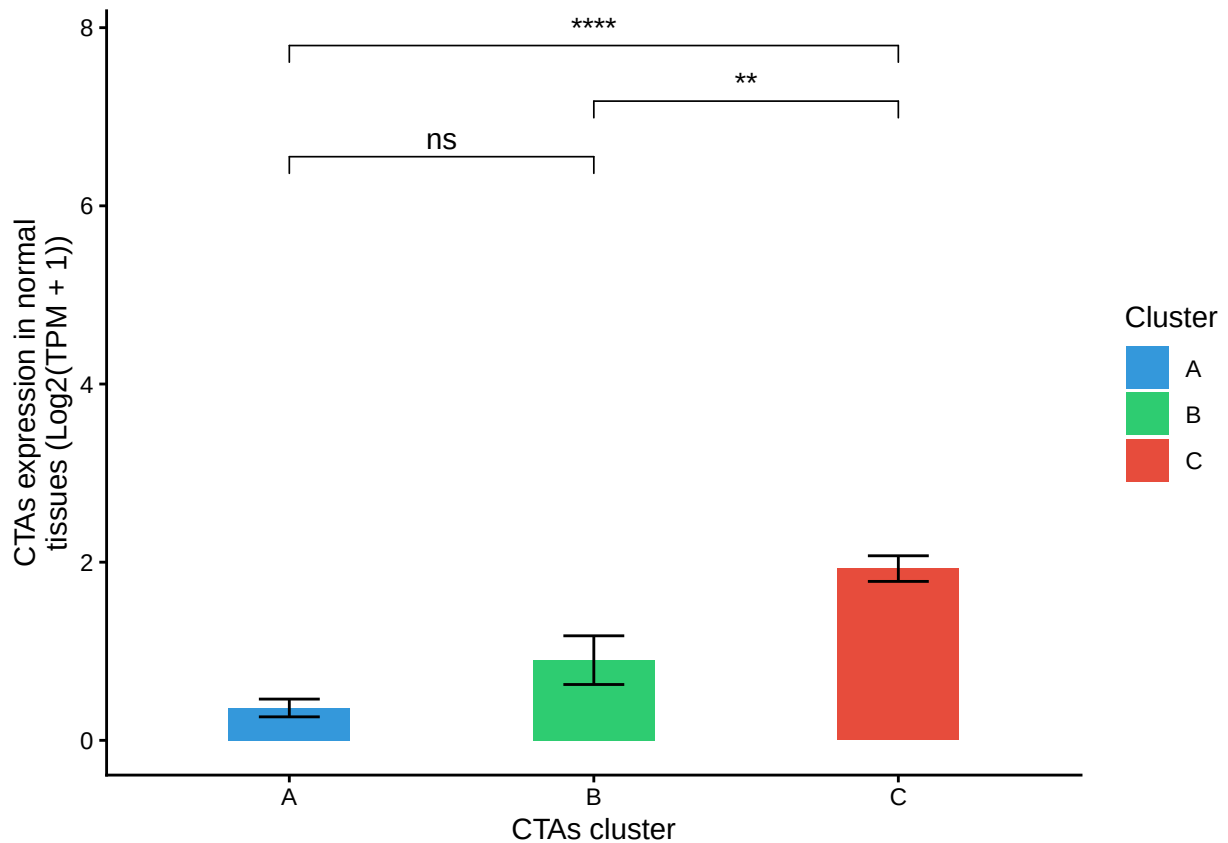


Figure 4: Barplot CTA expression per cluster

```
# ggsave('../results/barplot.svg', plot = plot_final, width  
# = 10, height = 10, units = 'cm')
```